

Cooperative Use of Unmanned Sea Surface and Micro Aerial Vehicles at Hurricane Wilma

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On Oct. 24, 2005, Hurricane Wilma, a category 5 storm, made landfall at Cape Romano, Florida. Three days later, the Center for Robot-Assisted Search and Rescue at the University of South Florida deployed an iSENYS helicopter and a prototype unmanned water surface vehicle, AEOS-1, to survey damage in parts of Marco Island, 14 km from landfall. The effort was the first known use of unmanned sea surface vehicles (USVs) for emergency response and established their suitability for the recovery phase of disaster management by detecting damage to seawalls and piers, locating submerged debris (moorings and handrails), and determining safe lanes for sea navigation. It provides a preliminary domain theory of postdisaster port and littoral inspection with unmanned

vehicles for use by the human-robot interaction community. It was also the first known demonstration of the strongly heterogeneous USV-micro aerial vehicle (MAV) team for any domain. The effort identified cooperative UAV-USV strategies and open issues for autonomous operations near structures. The effort showed that the MAV provided a much-needed external view for situation awareness and provided spotting for areas to be inspected. Concepts of operations for USV damage inspection and USV-MAV cooperation emerged, including a formula for computing the human-robot ratio: $N_h = (2 \times N_v) + 1$, where N_h is the number of humans and N_v is the number of vehicles. The outstanding research issues span three areas: challenges for USVs operating near littoral structures, general issues for USV-MAV cooperation, and new applications. It is expected that the lessons learned will be transferrable to defense and homeland safety and security applications, such as port security, and other phases of emergency response, including rescue.

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1. INTRODUCTION

On Oct. 27, 2005, three days after Hurricane Wilma, a category 5 storm, made landfall at Cape Romano, Florida, the Center for Robot-Assisted Search and Rescue (CRASAR) at the University of South Florida was invited by the Marco Island Yacht Club (MIYC), on Marco Island, Florida, approximately 14 km away, to investigate damage to their building and seawall using experimental unmanned vehicle technology. Figure 1 shows a map of the Marco Island-Cape Romano area, which is located on the west shore of Florida, near Naples. Although the damage to the

west coast of Florida fortunately turned out to be minor, the deployment serves as a field study of how unmanned vehicles, especially sea surface vehicles, can be used for disaster response and under what operating conditions. Though an actual deployment is not a reproducible, experimentally controlled event, the ethnographic observations provide valuable insights into concepts of operations (CONOPS) for unmanned surface vehicles and future research directions.

This article describes the deployment of an unmanned sea surface vehicle (USV) and a rotary-wing micro aerial vehicle (MAV) to assist with the disaster assessment and the lessons learned. USVs are also known as guided surface vehicles or autonomous surface vehicles. MAVs are a type of unmanned aerial vehicle (UAV) that can be carried in backpacks and are generally limited to 2 m or less in wing span. Fixed-wing and rotary-wing MAVs had been first introduced to emergency response two months earlier at Hurricane Katrina (Murphy, 2006; Murphy, Griffin, Stover, & Pratt, 2006), and post-Katrina follow-up work has detailed the CONOPS for MAVs flying near urban structures (Pratt, Murphy, Stover, & Griffin, 2006). What distinguishes Hurricane Wilma for robotics is that it is the first known deployment of a USV for emergency response and also the first known application of a cooperative USV-MAV team.

The relevancy of this article is not limited to the disaster response domain; it provides insights into possible CONOPS for unmanned systems for safety and security activities such as port security but also for general littoral warfare (Castelin & Bernstein, 2004). Littoral warfare is a much older topic than homeland security and provides an interesting opportunity for cross fertilization. The domain of disaster response shares two of the three characteristics of littoral warfare described by Castelin and

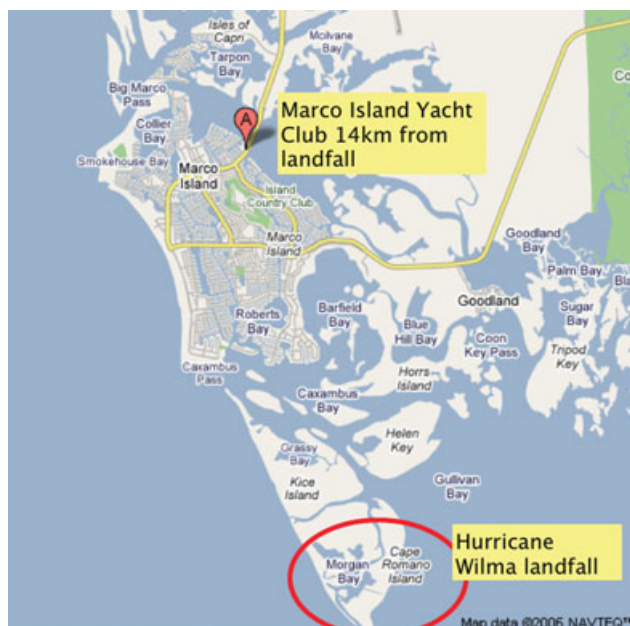


Figure 1. Map view of area of deployment (A) on Marco Island, Florida, and proximity to Cape Romano (map courtesy of Google).

Bernstein (2004). The littoral environment is complex, with access to the water and structures exacerbated by the density of natural (shores, sandbars, channels, currents, etc.) and man-made (bridges, piers, docks, boats, barges, navigation markers, etc.) obstacles. Both warfare and disaster management would benefit from faster operations, presumably through autonomous operations. The only characteristic that littoral warfare and disaster response does not share is the need for clandestine operations.

The article also helps identify general research issues for coordination of teams of highly heterogeneous robots. The bulk of the literature in heterogeneous teams, which is too large to cite here, have addressed either “weakly” heterogeneous teams in practice or “strongly” heterogeneous teams only through simulation. In weak heterogeneity, the individual members are variations of a single modality, usually ground robots. Heterogeneity stems from different sensor payloads or effectors on the team members. As a result, research with weakly heterogeneous teams is unlikely to transfer to coordination of strongly heterogeneous teams consisting of unmanned surface and aerial vehicles, where the individual control regimes and human–robot interaction (HRI) characteristics are significantly different.

This article greatly expands the preliminary video (Murphy, Steimle, Cullins, Pratt, & Griffin, 2006a) and report (Murphy, Steimle, Cullins, Pratt, & Griffin, 2006b) on the deployment. It first discusses the related work in rescue robots, heterogeneous USV–UAV teams, and general USVs and MAVs in Section 2. Next in Section 3 it introduces the equipment used, namely the AEOS-1 prototype USV designed by Eric Steimle for general environmental science research and the T-Rex platform modified by Chandler Griffin of iSENSYS, a corporate member of the National Science Foundation (NSF) industry/university cooperative research center on Safety Security Rescue Research Center affiliated with CRASAR. Section 4 describes two of the three missions conducted: a solo inspection of the MIYC seawall and the SR951 bridge and fishing pier by the USV and a cooperative USV–MAV inspection of the SR951 bridge and fishing pier. The USV and MAV were manually teleoperated, and the Wilma effort suggests that the shared control form of autonomy is desirable; the open research questions for semiautonomy and reducing the human–robot ratio are cataloged in Section 5. The article concludes in Section 6 that both USV and MAV technologies are useful for

disaster assessment and that MAV is especially helpful for USV control.

2. RELATED WORK

The work related to the use of unmanned systems at Hurricane Wilma can be divided into four categories. First, the work at Hurricane Wilma is unique because it is the first USV robot deployment for search and rescue; it joins unmanned ground vehicles (UGVs) and UAVs. Second, the work is the first known instance of any USV–UAV team, though there has been theoretical exploration of such teams. Third, the USV itself is similar to two other USV designs, though those platforms are not intended for response work. And finally, the rotary-wing MAV used at Wilma is related to work with a fixed-wing MAV designed for interior spaces.

The use of unmanned systems for rescue is still experimental, and the deployment reported in this article is unique. Ground robots were deployed by CRASAR at the 2001 World Trade Center disaster under the invitation of the State of New York (Murphy, 2004), the 2005 La Conchita, California, mudslide under the invitation of the Los Angeles County and Ventura County Fire departments (Murphy & Stover, 2006), and Florida Task Force 3 at Hurricane Katrina in 2005. MAVs were deployed by CRASAR during Hurricane Katrina through the Florida State Emergency Response Team (Murphy, 2006) and by defense contractors working through the U.S. Northern Command. Larger UAVs, such as the Predator, which require airspace coordination, were flown by at least two state Army National Guards at Hurricane Dennis (Florida) and Hurricane Rita (Florida, Texas). In addition to the actual responses, MAVs were used by CRASAR for structural inspection of damage from Katrina to multistory commercial structures along the Mississippi Gulf Coast by arrangement with the landowners (Murphy et al., 2006; Pratt et al., 2006). Goodrich et al. (Goodrich, Morse, Gerhardt, Cooper, Quigley et al., 2008) address the use of MAVs for wilderness search and rescue, though not for an actual event. That work captures the overall CONOPs for the overall strategic enterprise, which has a much higher degree of communications between tactical teams and the incident commander. This article concentrates on the tactical operation and constraints for USVs and for USV–UAV coordination under disaster conditions and contributes an understanding of the platform and software needs and the human–robot

ratio. There is no record of USVs being used for disaster response.

The literature examined did not yield any examples of fielded USV-UAV teams in general. Meyrowitz, Blidberg, and Michelson (1996) discuss the need for such partnerships. Dias et al. considered the issue of how to control heterogeneous teams including UAVs and USVs (Dias, Fraga, Gomes, Goncalves, Pereira et al., 2005). However, that work has produced command and control software tested in simulation and not implemented on real vehicles. Working on software, Yavnai (1994), in a similar effort, also considered UAV and USV cooperation from a theoretical standpoint.

There is some relevant work in small, easily portable USVs. Curcio, Leonard, and Patrikalakis (2005a) at the Massachusetts Institute of Technology (MIT) recently developed a fleet of four USVs based on small kayaks to conduct studies with cooperative autonomy with each other and with an autonomous underwater vehicle (Curcio, Leonard, Vaganay, Patrikalakis, Bahr et al., 2005b). The motivation was ease of use and reduced cost. Manley (1997) designed a USV with payload and size constraints similar to those of the AEOS-1. Like the AEOS-1, the motivation for the small size was to enable it to be transported by two people.

In terms of MAVs, the use of the rotary-wing MAV near structures is most similar to work by Green and Oh (2006) with a fixed-wing MAV. They designed a fixed-wing MAV for operations in interior spaces such as warehouses and caves. These interior environments present the same collision avoidance challenges as operations near buildings, bridges, and so on. However, interior environments are less challenging for vehicle control as there is less possibility of gusts and unpredictable wind effects.

3. EQUIPMENT

The equipment used at Hurricane Wilma consisted of the two unmanned systems and the support gear, described below. Both vehicles were teleoperated. The equipment was transported in two sport utility vehicles (SUVs) and a small trailer.

3.1. USV: AEOS-1

The USV system deployed at Hurricane Wilma consists of the AEOS-1 platform with a DIDSON acoustic

camera payload for below-the-waterline (BWL) surveys and a video camera mounted on a pan/tilt mast. There were two operator control stations, one for the PILOT, who used line of sight (LOS) and the view from the above-the-waterline (AWL) camera to teleoperate the USV, and one for the BWL MISSION SPECIALIST to control, observe, and record the output from the DIDSON. The PILOT or the support engineer served as the AWL MISSION SPECIALIST and controlled, observed, and recorded the output from the camera.

The AEOS-1 was designed to handle a variety of instrumentation with payloads up to 225 lb in up to 3-ft seas. As shown in Figure 2, it is a catamaran design with two polyethylene pontoons connected with



(a)



(b)

Figure 2. Unmanned surface vehicle: the AEOS-1 platform (a) and close-up of DIDSON acoustic camera payload (b).

a central T-shaped chassis that supports the instrumentation, power, communications, and control system. The bracketing and mounting system allows for the center of gravity and trim to be easily adjusted for each payload configuration prior to deployment. This optimizes performance and enhances safety during deployments. The footprint is 1.9 m long and 1.2 m wide, with a draft of 0.3 m, and it can be transported by two operators. An onboard wireless interface with additional video transmitters gives control of instrumentation deployed with the USV and is housed in a water-resistant enclosure. An electronic speed control is used to control the propulsion. The propulsion is a twin-screw design providing up to 160 lb of thrust, yielding a maximum speed of 4 kn with extremely high maneuverability. The power for the motors is supplied by Absorbed Glass Mat lead acid batteries. However the 350 lb of buoyancy offered by the pontoons allows for additional batteries to be used for propulsion and instrumentation, thus meeting the demands required by the user. When necessary a small generator is mounted to supply supplemental power to instrumentation.

The USV is engineered for autonomous and tele-operated control. A standard radio controller can be used to manually drive the vehicle. The computer interface uses a standard IEEE 802.11G wireless network to communicate to the electronic speed control. A graphical user interface (GUI) is used to monitor battery voltages and temperatures and to interface the joystick with the vehicle. A real-time video feed is used to help assist with piloting the vehicle for object avoidance and to maneuver in confined space. The video is transmitted through a separate dedicated wireless network (at 0.9, 1.2, and 2.4 GHz) to minimize bandwidth issues. The surface vehicle employs commercially available tracking software, as well as a custom software package developed at the University of South Florida. From the onset the AEOS-1 was designed to incorporate commercially available components whenever possible. This has kept the cost low and allows repairs to be made in the field with a quick trip to local merchants.

The DIDSON is a dual-frequency identification sonar that gives near-video-quality images for underwater identification and inspection of objects in turbid environments. The DIDSON used in this deployment operates at 1.1 and 1.8 MHz. This provides images from 1 to 30 m. The DIDSON has a field of view (FOV) of 29 deg, providing a viewing cross section that is half the distance in the FOV. If the oper-

ator is looking 5 m downrange, he will get a 10-m-wide image. The FOV and the ability to update the image at up to 10 frames per second allow for real-time inspection in a highly cluttered environment. Also by implementing some simple algorithms, obstacle height can be determined from the length of the acoustic shadow, the distance from the sonar, the sonar angle, and height. A trained specialist operates the DIDSON through a laptop that interfaces with the DIDSON through an Ethernet port on the top-side electronics box. Display and control software are designed to run on a standard Windows-based personal computer or laptop. The sonar images can be viewed using the computer interface or through an NTSC video system.

3.2. MAV: T-Rex

The MAV was a Like90 (now iSENSYS, Inc.) T-Rex miniature helicopter (Figure 3). The T-Rex platform is battery powered, which simplifies logistics and transportation on commercial flights. The helicopter carried a miniature pan-and-tilt visible light camera made by Lockheed Martin and also transmitted real-time video to a camcorder. It can hover in winds up to 35 mph.

The helicopter motion and camera position were handled through radio control by a two-person team of a PILOT and MISSION SPECIALIST. The PILOT used a handheld, two-joystick Futaba radio control (RC) controller operating on a 72-MHz radio frequency. The MISSION SPECIALIST also used a Futaba RC controller but on a different channel. The video



Figure 3. MAV: T-Rex.

was transmitted back on a 2.4-GHz frequency and recorded on a camcorder, which also served as the display device.

4. MISSIONS AND OBSERVATIONS

The deployment consisted of a series of four missions, three planned and one opportunistic. The MAV was first tasked to check the MIYC roof for damage that was not visible from the ground (Figure 4). As MAV structural inspection has been reported elsewhere (see Murphy, 2006) and adds no new insights, this mission will not be discussed here. Next, the USV was tasked to determine the extent and severity of the damage to the seawall bounding the MIYC dock (Figure 5). After finishing the inspection of the MIYC property, it was noticed that a span was missing from the fishing pier below the nearby SR951 bridge. The USV and MAV were deployed to concurrently, but independently, inspect the bridge and the damaged fishing pier. However, the concurrent inspection led to experimentation with cooperative maneuvers.

Observational data were collected from the missions and then analyzed to extract the lessons learned, described below, creating a descriptive field study. The observational data consisted of (i) videotapes and (ii) field notes of the robot and HRI, (iii) feedback on site with harbor and fire rescue personnel, and (iv) an outbrief after the deployment. The analysis of the data forms a descriptive field study. Descriptive field studies are commonly used in



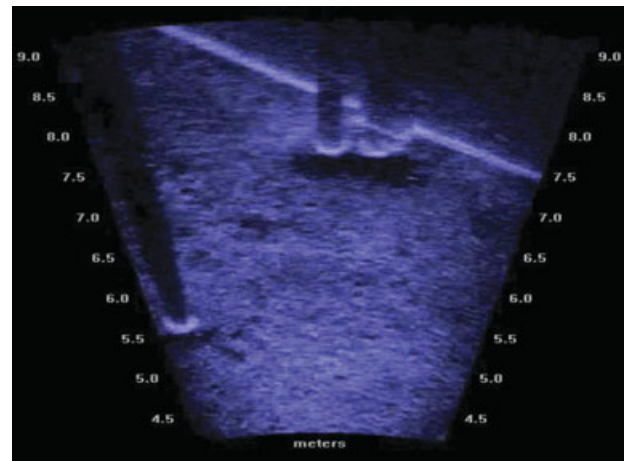
Figure 4. View of the MIYC; note the lack of access on the sides and dense landscaping.



(a)



(b)



(c)

Figure 5. USV inspection of MIYC seawall: external view (a), view from camera (AWL) (b), and view from USV DIDSON of seawall pilings (BWL) (c).

psychology, human factors, and artificial intelligence to establish the domain theory for a new application. The domain theory generated by the field study is the first step in formal research: it captures novel uses that have not yet been defined by the experts and provides an understanding of the conditions that must be reproduced for future experimentation. The analysis of the domain theory compared to the current state of technology suggests what might have been useful and therefore what the open research issues are. Field studies have three disadvantages: they are not reproducible (i.e., every disaster is different), there is no ground truth beyond expert assessment (i.e., Did the robots detect what was there?), and the analysis is subjective. However, field studies remain the primary mechanism for specifying a novel domain and posing research issues to be formally explored.

4.1. USV Seawall and Bridge Inspection

The second and third missions were to inspect the MIYC seawall and the SR951 bridge and fishing pier. Both structures would have normally required a boat and a diver. Although the seawall on the east side facing the inlet appeared damaged from the storm surge, the wooden dock prevented any viewing from AWL. In addition, significant damage to any of these structures could be BWL, which would normally require a diver to inspect.

4.1.1. Objectives

The research objectives of the seawall and bridge inspection by a USV concerned the payload, the platform, and the CONOPs. The first objective was for the team sonar expert, Cullins, to determine whether the DIDSON underwater acoustic camera was sufficient to detect structural damage even in high turbidity. The next goal was to establish whether the vehicle had sufficient maneuverability and reliable wireless communications to warrant future deployments as an emergency response asset. Finally, the mission was expected to provide feedback on general deployment practices and best practices.

4.1.2. Deployments

The USV team consisted of the Pilot, Eric Steimle, the sonar MISSION SPECIALIST, Charlie Cullers, and Mike Hall, who assisted as the MISSION SPECIALIST for the AWL camera or took turns as PILOT. The USV was

driven primarily LOS by the PILOT, with the extra person serving as the AWL MISSION SPECIALIST until it reached the littoral structure of interest. Then the PILOT would have to swap to using the camera to direct the vehicle. The two would move around and “walk” with the USV to maintain the best LOS and to make sure that no unsafe situation was evolving. The BWL MISSION SPECIALIST was stationary and remained with a dedicated laptop under the shaded entryway of the MIYC.

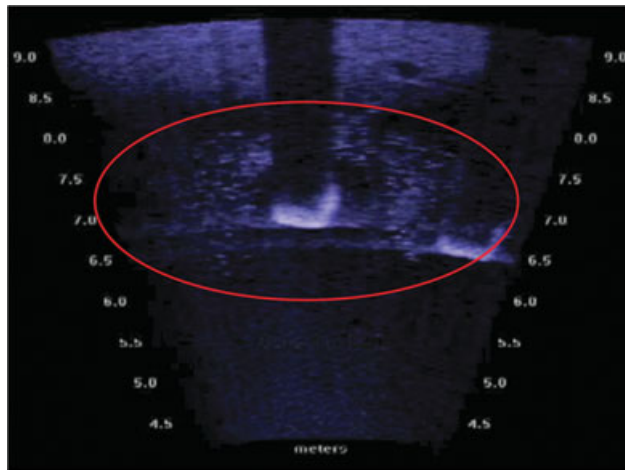
The USV was sailed twice. In the first sail, the AEOS-1 started in the Marco Island marina (with permission from the dockmaster) and circled around the MIYC and then out into the inlet and to the seawall. While operating in the marina, the USV found the vent cap from the roof. The second sail was to the SR951 bridge. During the second sail, the USV was beached and examined to make sure that all was still functioning.

4.1.3. Lessons Learned

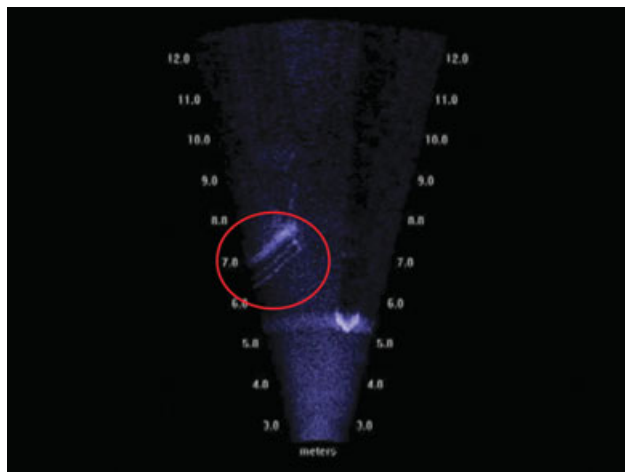
The USV seawall inspection mission met the research objectives. The two operational areas, the seawall and bridge, provided two different littoral environments, and the DIDSON performed better than expected. The vehicle was demonstrated to have sufficiently high mobility to work underneath small docks as well as enough power to operate around bridges in significant current. The mission also provided feedback on the deployment practices; these are described in the Open Research Questions (Section 5).

The missions showed the efficacy of the DIDSON camera in realistic conditions. The small size of the USV allowed the DIDSON acoustic camera to be placed close to littoral structures. Figure 6 shows two sample outputs. Figure 6(a) shows a piling for the fishing pier, with the resolution high enough to show a school of bait fish, each smaller than the palm of a hand. A set of railings from the missing section of the fishing pier was found, despite the muck and turbidity of the sea floor, as shown in Figure 6(b). The only limitation was the lack of panning head; the DIDSON was mounted on a tilting, but not panning, sensor head.

Possibly the most significant lesson learned was that the USV offered at least three advantages over manned surface vehicles and unmanned underwater vehicles (UUVs). The overall small size allows the USV to be more deployable in disaster conditions and to reach places that manned boats cannot



(a)



(b)

Figure 6. Pictures from DIDSON: school of small fish (a) and railing from missing section of fishing pier, most likely submerged in muck (b).

reach. UUVs have been proposed for inspection of littoral structures (Ohata, Ishii, Sakai, Tanaka, & Ura, 2005), identifying the need for such a vehicle to operate in 200 ft of water or less. The Wilma deployment showed that a USV can meet the desires for the UUV while avoiding many of the disadvantages. In particular, the USV can see AWL as well as BWL, plus it does not suffer from the limitations of underwater wireless communications, which interfere with control and real-time observation.

First, the small size allowed it to be deployed without a boat ramp and to be used under docks and piers that a manned system could not reach. The

AEOS-1 was light enough to be lowered by hand into the water and to drive up to the beach. Although the small size was suitable for deployment, it is desirable to have a slightly smaller platform in order to reduce the logistics trail. The system was transported in a trailer; trailers might not be able to traverse damaged areas, and therefore the goal is to have the system be transportable by a SUV.

A second advantage is the ability to examine the structures simultaneously AWL and BWL. The AEOS-1 was able to locate sections of the seawall corresponding to where the dock above had been damaged by the storm surge and confirm that the damage was superficial. The damage did not extend into the seawall or its foundation.

The third advantage is that the USV can be easily controlled in real time and transmit video imagery over a wireless network. In contrast to USVs, UUVs must either be tethered to a transmitter buoy AWL or rely on an underwater acoustic modem. Tethering introduces issues of tangling. Underwater modems have an effective throughput rate of 17.8 Kb/s, not sufficient for gathering real-time imagery or driving a vehicle in close proximity to structures.

4.2. MAV-USV Cooperative Bridge Inspection

The third mission was originally intended to explore the complementary nature of MAV and USV inspection of littoral structures (Figure 7). However, it evolved into an investigation of cooperation and coordination between the two platforms.



Figure 7. AEOS-1 and T-Rex in close formation.

4.2.1. Research Objectives

The research objective of the USV–MAV exercise originally was to identify the requirements for bridge inspection by either USVs or MAVs, assumed to be working independently of each other. Our prior experience at Hurricane Charley (Murphy, Stover, & Choset, 2005) had established that postdisaster bridge and transportation infrastructure inspection is a potential task for unmanned systems. The work earlier in the day with the MIYC seawall suggested that the USV would be able to adequately survey the footing BWL. The idea was that a MAV and a USV could concurrently inspect a bridge, providing completeness. A question was whether the small helicopter would be able to operate in the wind conditions around the bridge and, if so, how much faster the MAV could work than the USV.

However, the objective changed once the USV and MAV began concurrent operations at the bridge. Because of the lateral perspective, it was very hard for the USV Pilot to see exactly where the vehicle was relative to the bridge and fishing pier. The MAV provided a very useful viewpoint for localization, and so the exercise expanded into an exploration of three issues in the USV–MAV coordination: *How close did the MAV need to fly to track the USV?*, *What viewpoints were preferred by the USV Pilot?*, and *What are the general issues of vehicle coordination and mixed-team processes?*

4.2.2. Deployment

The USV–MAV cooperative inspection exercise used five team members: the two-person USV team (PILOT, MISSION SPECIALIST), the two-person MAV team (PILOT, MISSION SPECIALIST), and a general FLIGHT DIRECTOR. The sixth person assisted as needed. Communications between the USV and MAV teams were done verbally and via radio. The USV team operated at the foot of the fishing pier, launching the USV from the beach. Their viewpoint was primarily parallel to the fishing pier and bridge, causing problems with depth perception. The MAV team stayed on the beach, also with a parallel vantage point.

The USV operated continuously for approximately 1 h, and the MAV made three flights of about 15 min each. The USV surface plan was loosely specified as to survey the fishing pier pilings, especially the damaged section. Later, the plan was extended to include cruising around the infrastructure, providing tracking challenges for the MAV. The MAV flight plan was identified and verbally rehearsed, emphasizing

keeping the MAV in the PILOT's LOS with no occlusions and where to land. The operation over water was a significant concern. The sun was at the MAV PILOT's back for the flight, but the sun was often in the MAV MISSION SPECIALIST's camera view, causing lens flaring and general optics difficulty.

4.2.3. Lessons Learned

The primary lesson learned was that the use of a MAV with a USV offers many advantages. Three advantages identified through the work at Hurricane Wilma are as follows: the MAV can (i) assist with relative positioning of the USV to the structure, (ii) enhance safe operations of the USV, and (iii) serve as a communications relay. Work such as that of Grocholsky, Keller, Kumar, and Pappas (2006) with UAVs assisting with localization of UGVs have noted similar advantages. As a result, we now prefer to field the USV with MAV support, rather than solo. It should be noted that MAVs can actively control the viewpoint of the USV to optimize tracking the USV for the structural and environmental conditions as it travels through complex structures; this is a distinct advantage over tethered blimps such as reported by Sevcik, Green, and Oh (2005).

The MAV provides an external, "bird's-eye," view of the USV. This view allows the USV PILOT to localize the vehicle for navigation relative to the structure, compensating for either the lack of global position system (GPS) or the GPS errors and the constricted LOS of the PILOT. Software such as Falconview (Georgia Tech Research Institute, 2007) is already commonly used in U.S. military MAVs to determine the GPS coordinates of objects in MAV imagery, and there have been significant efforts to increase the accuracy of such systems (Johnston, 2006; Quigley, Goodrich, Griffiths, Eldredge, & Beard, 2005; Redding, McLain, Beard, & Taylor, 2006).

However, these packages allow the MAV to map objects but do not necessarily help the USV navigate. Because the USV is operating in a GPS-denied area, being provided GPS coordinates from the MAV does not help it navigate to specific damage or ensure coverage of a search within the structure. In this case, the visual view of the USV provided by the MAV is likely to be extremely important for relative localization. Relative localization can be either coarse, where the issue is getting the vehicle to a general neighborhood of interest, or fine, where the vehicle must move in areas smaller than the error in GPS. As



(a)



(b)

Figure 8. Two views of the missing section from the fishing pier: from the staging area (a) and from the MAV, which clearly shows the missing section (b).

an example of coarse relative localization, consider Figure 8(a). The figure shows a view of the damaged fishing pier from the point of operations; the damage cannot be seen and even if it was, the GPS error might guide the robot to an adjacent piling. However, the view from the MAV shown in Figure 8(b) clearly shows the missing section and the relative position of the USV. With this view from the MAV, the USV PILOT can drive the robot to the desired location. Fine relative localization is also needed to compensate for the PILOT's constricted LOS through the USV cameras. Figure 9 shows the density of the structures



Figure 9. Aerial view from the MAV of the USV after it passed under the fishing pier and continued within the infrastructure of the SR951 bridge, posing tracking problems.

that may be encountered. The repetitive placing and lack of perceptual distinctiveness could lead to an autonomous USV or a USV PILOT confusing one piling with another as it was tried to survey the complete area or built a local map. Because the robot is operating within the margin of GPS error, having a GPS signal would not help. The external MAV view would allow the PILOT (or a software agent) to maintain an understanding of where the MAV actually is and where it has been.

The bird's-eye view provided by the MAV can also help with general safety and control. It allows the team to see whether there are other watercraft or large debris (e.g., fallen trees, pieces of docks or houses) in the area that might move into the path of the USV. The view can also identify the direction of waves that might interfere with USV performance.

Another advantage of the MAV, though this was not implemented at Wilma, is that it can serve as a communications relay. The use of MAVs as a communications relay is well known (see Ayyagari, Harrang, & Ray, 1996; Chen-Mou, Pai-Hsiang, Kung, & Vlah, 2007; Palat, Annamalai, & Reed, 2005; and Pengcheng, Kai, & Swindlehurst, 2006, for examples). As will be discussed in Section 5.1.3, loss of wireless communications is a major concern. The MAV provides a much more favorable node, in both altitude and ability to move, and increases the overall reliability of the network.

5. OPEN RESEARCH QUESTIONS

Taken altogether, the experiences at Hurricane Wilma lead to several research questions. These can be loosely grouped into four areas: challenges for USVs operating near littoral structures, challenges for MAVs operating near urban and littoral structures, general issues for USV–MAV cooperation, and new applications. The challenges for USVs operating near littoral structures are AWL/BWL SLAM (simultaneous localize and map), situation awareness given the differences between AWL/BWL, intermittent GPS and wireless communications signals, and general impact of the maritime environment.

5.1. USV Challenges

The challenges for the use of unmanned surface vehicles in cluttered littoral regions can be grouped into four broad categories. One is how these robots will SLAM both AWL and BWL. Another is how an operator or user will maintain situation awareness given the potential disparity in FOV BWL and AWL. The interference of radio signals, especially GPS and communications, in cluttered littoral regions such as bridges is another issue. Finally, the maritime environment itself presents a challenge for sensing and control.

5.1.1. AWL/BWL SLAM

Figures 5(b) and 5(c) illustrate some of the challenges for SLAM of a USV. Unlike UGV SLAM or UUV SLAM, USV mapping has to take into account features or data above [Figure 5(b)] and below [Figure 5(c)] the waterline. The data from above and below the waterline are likely to be of different resolutions and accuracy. It seems unlikely that a single algorithm will be able to merge the uncertainty from the two regions, and there appears to be no work on accurately fusing separate underwater and above-water maps.

5.1.2. Above/Below Situation Awareness

Situation awareness is a broad term. Two aspects of situation awareness that an operator or software agent will have to maintain are about (i) the internal and external state of the USV (e.g., the health and pose of the robot relative to the world) and (ii) interpretation of sensory data. Of these two, the inter-

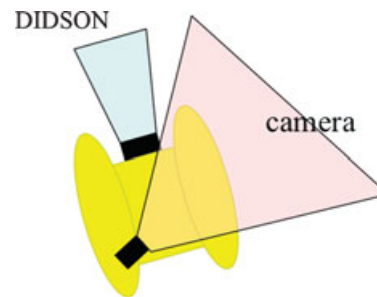


Figure 10. Example of a challenge to operator situation awareness, where one imaging sensor (DIDSON) can be oriented differently from another sensor (camera).

pretation of sensory data is the harder, less-examined research issue. Figure 10 illustrates the problem of interpreting sensory data: how an agent will make sense of imagery coming from different viewpoints as well as above and below the waterline. Whereas Figure 5 shows the qualitative difference in imagery above and below the waterline, Figure 10 shows that the images are not necessarily connected. This decoupling of viewpoints suggests that it will be very hard for an operator to form a single, correct understanding of the operational environment of the USV without significant help from intelligent software. Without such an understanding, the operator may miss indications of structural problems. Also, the operator may try to direct the robot to places it cannot safely reach, either damaging the robot or creating frustration when the robot, operating under semiautonomous guarded motion, does not obey.

5.1.3. Intermittent GPS and Wireless Communications

As shown by the performance at Wilma, urban structures such as bridges interfere with two types of important signals: GPS and wireless communications. The loss of GPS near structures presents navigation, especially localization, problems. Once near or under a bridge, a USV will have to resort to some other localization method. Given wave action and currents, it is unlikely that an inertial measuring unit (IMU) will be sufficient.

Wireless communications is needed for real-time control and observations, but in practice, wireless networks remain hard to set up and have limited bandwidth. Sugiyama et al. (Sugiyama, Tsujioka, & Murata, 2006) pointed out the difficulty of

maintaining quality of service in networks for general urban search and rescue (USAR). The Wilma deployment did not show loss of signal, with the USV using commercial 802.11 wireless, while the MAV used 72 MHz and 2.4 GHz. However, it is estimated that 2 of the 6 h spent in the field were spent setting up the wireless communications such that the USV and MAV did not interfere with each other. Bandwidth limitations on the 802.11 network were particularly noticeable with the USV because there were two video feeds: from the AWL camera and the BWL acoustic camera.

5.1.4. Maritime Environment

The environment itself poses challenges for autonomous operation of USVs. One set of challenges are atmospheric effects such as fog, which can interfere with visual cameras. Changes in lighting, from bright sun to shadows under bridges and docks, also are a concern. These concerns suggest the need for infrared or laser illumination. The sea itself is a problem, with waves, currents, and tides. Waves not only change the pose of the USV in ways that are difficult to model but may splash water and debris onto sensors. AWL sensors will have to be water resistant and self-cleaning. Waves also present challenges to range sensing, because the surface is not flat or horizontal or even stationary. A naive range algorithm might misinterpret waves as obstacles with low profiles such as boats. The impact of tides is another problem, given that the USV is operating in littoral regions. The ability to reason about changing depths is important and adds to the challenges of SLAM.

5.2. USV–MAV Cooperation

Independently of the demands of USVs and MAVs for operations in littoral regions, the work at Hurricane Wilma helped identify at least some of the general challenges in USV–MAV cooperation. Although MAVs need some capabilities for operations in cluttered littoral regions, additional capabilities such as tracking and reasoning are needed in order to maintain a view of the USV. HRI is important for interacting with the MAV and USV, but the human–robot team presents additional HRI challenges. Finally, another challenge is introduced by beyond-LOS operations (BLOS). In theory, a MAV can extend the operation of the USV BLOS; however, it is not clear how this configuration can be maintained safely.

5.2.1. MAV Tracking USV Enhancements

Although a MAV and a USV can cooperatively inspect littoral structural damage, it appears that an important function of the MAV in a USV–MAV team will be to track and spot the USV as it moves near and under bridges and docks. The mechanics of tracking is similar to the work reported by Kanchanavally, Ordonez, and Layne (2004) and Grocholsky et al. (2006), which provide a preliminary consideration of the problem of flying a UAV in difficult terrain to follow a ground target. The experience at Hurricane Wilma suggests that in order to fulfill this function, the MAV must have at least three competencies: avoidance, tracking, and viewpoint reasoning.

First, the MAV must be able to avoid collisions with these structures. However, two points warrant emphasis. The MAV is likely to be operating close to the structure to provide the best viewpoint, leaving little room for error. The unpredictable wind conditions near complex structures place additional demands of avoidance.

Second, it would be desirable for the MAV to be able to predict where the USV is going in order to track it. One approach is a traditional tracking algorithm that predicts the path of the USV based on its historical trajectory. However, because the USV is operating in clutter, the trajectory may change radically; for example, the USV may be moving parallel to a bridge and then turn to go under the bridge and circle the footings. Another approach of interest is to exploit the possibilities of joint communication between the team: that the USV can tell the MAV where it is going and what its current control commands are.

Third, the MAV should maximize both the time the USV is in view and the value of the viewpoint. The MAV may not be able to maintain a continuous view of the USV as it goes through complex structures such as bridge supports. Figure 9 shows such a case, in which the USV is operating within the envelope of the SR951 bridge where the MAV cannot safely maneuver. Therefore, it is desirable for the MAV to be able to reason about occlusions and position itself at a hovering location to maximize viewing coverage of the USV. One possibility is to use a priori information about the structure to help reason about where the MAV should place itself. There will likely be a set of possible hovering positions, and each will have a different value. For example, higher viewpoints are often more informative in determining relative distance between two objects than a lower angle.

5.2.2. HRI

The appropriate HRI for USV–MAV teams is an open research question. Because rescue robotics is a remote-presence domain, where the robot is an extension of studies by Birk & Carpin (2006), who argue that rescue robotics is critical for understanding human-in-the-loop autonomy. There are at least three issues of interest: What is the optimal human–robot ratio?, What are the roles for each human?, and How can the general team performance be facilitated? Although only rudimentary ethnographic notes were collected at Wilma, they do provide an understanding of what might be considered the state of the practice, which is represented by Eqs. (1) and (3). The question of which of these equations is more correct from a human factors standpoint is an important, open research question that should be verified experimentally in order to establish the baseline of teleoperations. Once the baseline ratio for safe teleoperation is established, experiments with adding autonomy should be conducted.

On the basis of the experiences at Wilma, combined with our related work at Hurricane Katrina (Murphy, 2006; Murphy et al., 2006; Pratt et al., 2006), we consider the appropriate human–robot ratio for a MAV–USV team to be with a five-person team. This would be a 5:2 ratio, with three human roles (PILOT, MISSION SPECIALIST, and FLIGHT DIRECTOR). As noted in Section 2, prior work has shown the efficacy of a 2:1 ratio for ground robots (Burke, Murphy, Coover, & Riddle, 2004) and a 3:1 ratio for MAVs (Murphy et al., 2006). These ratios can be generalized by the following formula, where the number of humans N_h currently needed to operate a cooperative team with N_v vehicles can be estimated as

$$N_h = (2 \times N_v) + 1. \quad (1)$$

The formula is derived as follows. One person is needed to assume the PILOT role for each vehicle, and one is needed to be a dedicated MISSION SPECIALIST. Whereas the USV used two MISSION SPECIALISTS, one for the BWL DIDSON sensor and another for the AWL camera, the USV PILOT often doubled as the AWL MISSION SPECIALIST, swapping between LOS and through-the-camera views. An additional person is needed to serve as the overall FLIGHT DIRECTOR.

However, this formula may be too conservative. A single FLIGHT DIRECTOR may not be sufficient. The FLIGHT DIRECTOR has traditionally used an unmedi-

ated perspective to keep up with the greater situation. This works well with LOS operations of the MAV and near operations of the USV. However, the MAV is helpful in providing the larger view of the USV work area, which is not available from the ground. This view of the USV from the MAV is mediated; that is, it appears on a computer screen and requires the observer to stop looking at the other surroundings (even with “see through” heads-up display glasses). It is not clear whether the FLIGHT DIRECTOR can safely alternate between the MAV bird’s-eye view and unmediated views without getting captured by one viewpoint and missing something important. Therefore, two FLIGHT DIRECTORS may be necessary.

The formula may be too conservative in terms of mission specialists. The Hurricane Wilma deployment frequently used two MISSION SPECIALISTS plus a Pilot, though the Pilot did often double as the AWL MISSION SPECIALIST when team members were doing other things. The USV presents the case in which the AWL and BWL payloads have quite different cognitive demands. The impact of diverse payloads p on the ratio could be captured with a modification of Eq. (1):

$$N_h = N_v + N_p + 1. \quad (2)$$

Facilitation of team processes is an open question. The experience at Hurricane Wilma and Hurricane Katrina (Murphy, 2006; Murphy et al., 2006; Pratt et al., 2006) suggests that all team members should be physically colocated so that the two PILOTS can easily talk with each other and coordinate vehicle activities.

5.2.3. Impact of Autonomy and Interfaces

Although the scope of this article is limited to findings from the field, some speculation as to the impact of autonomy and interfaces on reducing the human–robot ratio given in Eq. (1) is appropriate.

Given the current advances in MAV autonomy, it may be possible to transfer some of these roles to the robot. With the addition of autonomous hovering, it may be possible to serially combine the MAV PILOT and MISSION SPECIALIST roles, where the PILOT navigates the MAV to an observation point, places the MAV in autonomous hover capability, and then begins serving as the MISSION SPECIALIST operating the payload. Future autonomy advances, such as autonomous takeoff and landing, may eliminate the

need for a human MAV PILOT altogether. In this scenario, a human PILOT could drive the USV to an appropriate location, launch the MAV, place it in hover mode, place the payload in “track the USV” mode, and then resume piloting the USV. However, the reliability of autonomy with the possibility of loss of GPS and radio signals must be considered. Furthermore, the issue must be considered of whether a human can assume control of a MAV quickly enough if a failure of autonomy occurs; this is known in human factors as the out-of-the-loop (OOTL) control problem. Regardless of advances in autonomy, the Federal Aviation Administration (FAA) may be unwilling to permit full autonomy of MAVs for operations in the U.S. national airspace.

Advances in USV autonomy may be helpful in reducing the human-robot ratio but require significant research. Unlike the MAV, which can be stationed away from bridges or other structures that block GPS signals, the USV is definitely operating in GPS- and radio-denied areas. Therefore, it must be able to localize itself using the environment, either AWL or BWL.

Certainly, appropriate user interfaces can reduce cognitive load. Improvements to the basic one window per payload and control function GUI on the USV are expected to reduce demand. In the structural inspection application described in this article, the most likely set of interfaces to consider improving are the USV payloads, with the hope of creating a unified AWL and BWL view of the environment so that a single person can review them. This does assume that such a display is possible and the person will be expert in interpreting both AWL and BWL structural cues. The nature of identifying interesting structural deviations or understanding a disaster is difficult and open ended; therefore fully autonomous detection is not expected to be possible in the near future. However, even if improved interfaces reduce cognitive load, the reduction may not translate into reduced manpower because of the OOTL control problem.

5.2.4. Safe BLOS Operations

Currently, USV and MAVs are operated in the LOS of the PILOT and FLIGHT DIRECTOR in order to be safe. In theory, the MAV serving as a spotter for the USV can extend the USV to BLOS, allowing the USV to be operated safely at a larger range. However, this introduces new HRI issues. One is, How can the FLIGHT DIRECTOR maintain the broad overall awareness

needed to conduct the activity safely? If the FLIGHT DIRECTOR begins using the MAV view to maintain situation awareness, that awareness is now mediated by the MAV viewpoint and will neglect the view of the team members, other assets, and so on. Another is, How can the USV PILOT role be enhanced by the MAV view? This is both an HCI problem (e.g., How to display the MAV and USV views?) and a workload problem (e.g., Can a PILOT reliably combine views and make correct decisions, particularly in cluttered environments?). Although autonomy may enable BLOS operations, autonomy does not guarantee that this will be the most reliable solution for all conditions; more research in HRI is warranted.

5.3. Additional Tasks and Applications

The work at Hurricane Wilma clearly establishes the utility of USVs, MAVs, and cooperative USV-MAV teams for emergency response. However, this is a new domain, so the specific tasks and related applications are not yet known. In addition to inspection of littoral structures, our work has identified at least two other uses of USV-MAV teams for emergency response.

One application suggested by the work at Wilma suggested that MAVs may be used to target areas to be sampled by the USV. For example, at Wilma, the MAV identified an unusual coloration in a portion of the inlet that was not visible from the ground [Figure 11(a)]. In addition, it was noted by the chief of one of the fire departments in the affected area that MAVs could be used to identify when marinas damaged by the hurricane such as in Figure 11(b) began leaking diesel fuel into the sensitive waterways, perhaps by initially seeing a slick and then using the USV to trace back to the source.

Another application is the inspection of the general transportation and shipping infrastructure after a disaster. After a hurricane, harbors and marinas can be cluttered with debris (such as the roof vents blown off the MIYC into the harbor), sunken boats, and so on. This requires any responders using boats to proceed cautiously, and the amount of debris can limit the size of the boat. The possibility of debris can, in turn, limit the use of boats to remove trapped victims or transport supplies. It can also impact the resumption of commercial operations, such as shipping. A USV-MAV team could assist waterborne responders and the harbor master in determining safe lanes of travel.



(a)



(b)

Figure 11. Two examples of possible USV–MAV applications: an unidentified plume in the water extending throughout the Marco Island inlet as seen from the MAV (a) and leaking diesel fuel lines at a nearby Isles of Capri marina damaged by Hurricane Wilma (b).

6. SUMMARY AND CONCLUSIONS

The fielding of a prototype AEOS-1 USV and an iSENSYS rotary-wing MAV at Marco Island, Florida, after Hurricane Wilma serves as a case study and an exploration of the immediate research needs for USV and USV–MAV cooperative operations in urban littoral environments. It suggests that USVs are better suited than UUVs for operating close to structures in very shallow littoral regions and that USVs will perform such missions better with a MAV partner. However, advances in sea and flight semiautonomy, situation awareness, wireless communications, track-

ing moving targets through cluttered environments, and HRI are needed to make USV–MAV technology tractable. Although this work has concentrated on the use of the USV and MAVs for disaster recovery, the lessons learned should be transferrable to safety and security applications, such as port security, and other phases of a disaster, such as rescue.

This article provides three fundamental contributions to robotics and coordinated operations of strongly heterogeneous teams, and it also contributes to the advancement of the field of search and rescue robotics. Specifically, the effort

- *is the first known deployment of a USV operating by itself or in a USV–MAV team for disaster recovery.* As a result, this work establishes their viability as an additional tool for emergency response and provides the first case study of their operation for the robotics community. Whereas this case study is a single event that is not reproducible, it is expected that the observations from Hurricane Wilma will be a foundation for future USV–MAV work.
- *is the first known example of a USV–MAV cooperative team outside of simulation.* The Wilma deployment suggests that cooperative USV–MAV teams are desirable for operations in littoral environments and provides a preliminary CONOPs, where the MAV assists with localization and situation awareness. The most significant problems may be the navigational autonomy and spatial reasoning needed for the MAV to maintain a view of the USV as it operates in cluttered urban littoral environments. In terms of HRI, the effort suggests that the human–vehicle ratio, or $N_h:N_v$, for a teleoperated heterogeneous team is

$$N_h = (2 \times N_v) + 1.$$

The Hurricane Wilma deployment identified specific USV–MAV research questions as well as general HRI issues. Two categories of open research questions are unique to USVs and USV–MAV teams operating in cluttered littoral or urban environments: enabling the MAV to provide situation awareness information for the USV and general USV localization and navigation near urban littoral structures. The general issue of the appropriate human–robot ratio and how that ratio can be reduced, or roles adapted, given advances in autonomy is equally important, though

not unique to this domain. Because these research questions are generated from an actual field deployment, they provide a de facto prioritization of research needs over the many other possible topics in autonomy and robotics.

Of the two categories of problems, enabling a MAV to contribute to the construction and maintenance of situation awareness may pose the most challenges. The deployment at Hurricane Wilma indicated that USV control and general situation awareness would be greatly improved through the use of the MAV as a “spotter.” To reliably operate as spotter, the MAV needs to be able to safely fly around and through complex, three-dimensional urban littoral structures such as bridges and piers. Certainly MAV flight autonomy is an existing topic with considerable research activity. What USV–MAV coordination adds to the flight autonomy challenge is the issue of maintaining the best viewpoint of the USV and environment to maintain situation awareness. This involves tracking of the USV, which might be temporarily occluded and may be changing its path, but also requires real-time spatial reasoning with a priori information that may have changed due to the disaster (e.g., the bridge has a partial collapse).

The missions and activities conducted at Hurricane Wilma are not an exhaustive list of what can be done with a USV, MAVs, and/or USV–MAV teams for emergency response; several other potential applications were identified. One application is for identifying hazardous material spills. A MAV, or longer duration fixed-wing UAV, can be used to detect plumes or disturbances in the water to be sampled by the USV. Without this bird’s-eye view, the USV might not sample those areas or, given the limited time, might not even be deployed in that area without some indication that it was needed. Another possible application is for the USV, MAV, or team to be used to assess commercial and recreational waterways and shipping lanes.

Under funding from the Office of Naval Research (ONR), we are constructing and programming a new USV, the SEA-RAI, specifically suited for close operations in urban littoral environments. Our research is also addressing expanding MAV autonomous collision avoidance, using the MAV to provide wireless communications relaying, and continuing to refine the HRI requirements. The 2006 hurricane season did not have any hurricanes make landfall in the United States, but the AEOS-1 and the new SEA-RAI USV and iSENSYS family of MAV remain available for use.

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